



Time: 5 minutes

Outcome: Trace Changes in Surplus & Policy Effects.

Difficulty: Intermediate

Changes in Surplus & Policy Effects

THE CORE IDEA

A price or policy change can transfer surplus between buyers and sellers and can destroy surplus by preventing mutually beneficial trades. A transfer changes who receives existing gains without changing their total. Deadweight loss is different: it represents gains from exchange that disappear because quantity moves away from the efficient level.

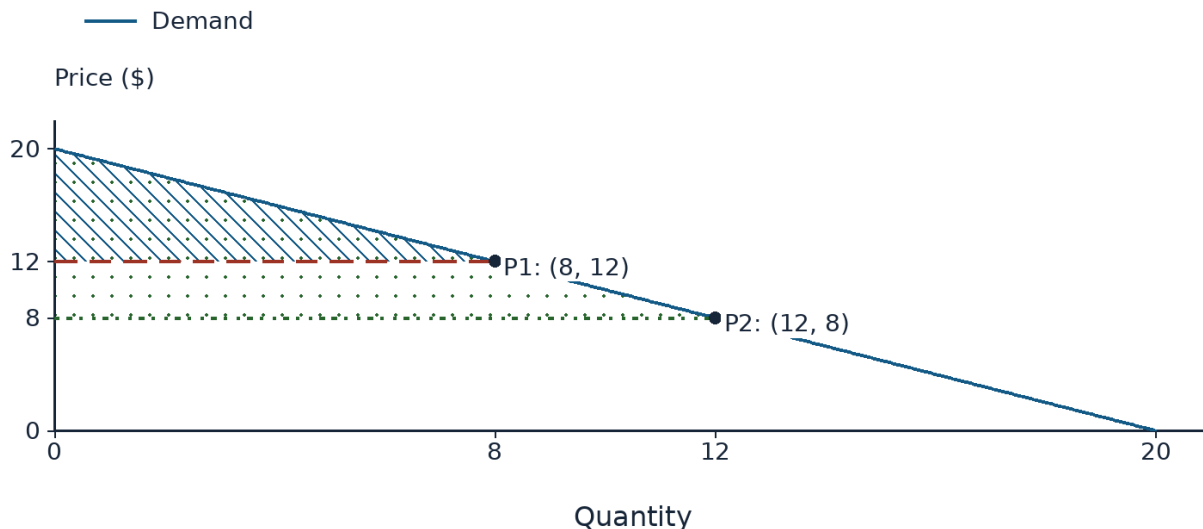
HOW TO RECOGNIZE IT

- Establish the initial consumer surplus, producer surplus, and total surplus before the change.
- Recalculate those areas after the price change or policy intervention.
- Separate transfers between buyers and sellers from surplus that disappears entirely.
- Deadweight loss represents mutually beneficial trades prevented by the change.

WATCH OUT

A demand curve and price are enough to calculate consumer surplus. A change in consumer surplus alone does not establish the change in producer surplus, total surplus, or deadweight loss.

WORKED EXAMPLE: MEASURING A CHANGE IN CONSUMER SURPLUS



At price $P_1 = \$12$, quantity is 8 and consumer surplus is $\frac{1}{2} \times 8 \times (\$20 - \$12) = \32 . At $P_2 = \$8$, quantity rises to 12 and consumer surplus becomes $\frac{1}{2} \times 12 \times (\$20 - \$8) = \72 . The price decrease raises consumer surplus by \$40. These demand-side areas do not identify a transfer or deadweight loss without supply-side information.

CHECK YOURSELF

Consumer surplus rises from \$32 to \$72 after a price decrease. What is the gain, and can this information alone establish the change in total surplus?